

ZURICH AP, Switzerland

WMO: 066700

Lat: 47.4797N

Lon: 8.5358E

Elev: 1416

StdP: 13.96

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 20.3	(d) 10.1	(e) 9.7	(f) 19.2	(g) 14.7	(h) 12.2	(i) 22.7	(j) 24.3	(k) 47.0	(l) 21.7	(m) 44.9	(n) 5.0	(o) 40	(p) 0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.0	(c) 87.5	(d) 68.2	(e) 83.9	(f) 66.7	(g) 80.6	(h) 65.7	(i) 69.7	(j) 83.2	(k) 68.1	(l) 80.6	(m) 66.6	(n) 77.9	(o) 6.0	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	97.8	74.2	63.9	93.6	72.5	62.6	89.5	71.0	34.4	83.6	33.1	81.0	32.0	78.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.4	(b) 15.4	(c) 13.0	(e) 9.9	(f) 92.1	(g) 5.2	(h) 3.1	(i) 6.2	(j) 94.3	(k) 3.1	(l) 96.1	(m) 0.2	(n) 97.8	(o) -3.6	(p) 100.1		
(a) 18.4	(b) 15.4	(c) 13.0	(e) 9.1	(f) 72.8	(g) 5.4	(h) 1.3	(i) 5.2	(j) 73.7	(k) 2.1	(l) 74.5	(m) -0.9	(n) 75.2	(o) -4.8	(p) 76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	33.6	35.3	42.6	49.4	57.1	63.7	66.9	66.1	58.4	50.8	40.8	35.0		
	DBStd	13.59	7.33	7.58	6.91	6.68	6.11	6.49	5.62	5.60	5.70	6.47	6.58	6.98		
	HDD50	2090	509	411	243	92	12	1	0	0	5	70	281	466		
	HDD65	5729	974	831	696	467	254	101	45	54	210	441	725	931		
	CDD50	2110	1	0	12	74	231	413	525	499	255	94	6	0		
	CDD65	275	0	0	0	0	8	63	105	88	11	0	0	0		
	CDH74	2818	0	0	0	17	152	678	1034	834	102	1	0	0		
	CDH80	834	0	0	0	1	20	197	335	274	7	0	0	0		
(14) Wind	WSAvg	5.0	5.6	5.7	6.0	5.4	5.2	4.8	4.7	4.1	4.4	4.4	4.6	5.4		
(15) Precipitation	PrecAvg	47.5	2.7	2.8	3.0	3.5	5.2	5.4	5.5	5.2	3.8	3.5	3.5	3.5		
	PrecMax	60.7	6.7	6.4	8.9	6.9	9.6	8.7	10.1	9.3	8.2	6.3	7.7	7.3		
	PrecMin	37.4	0.4	0.8	0.9	0.5	1.2	2.0	2.1	0.7	1.8	0.3	0.0	0.0		
	PrecStd	6.1	1.7	1.5	1.7	1.7	2.3	1.7	2.1	2.0	1.6	1.6	1.9	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	58.2	68.0	76.5	82.6	90.1	92.2	92.6	80.8	71.6	61.4	53.8		
		MCWB	48.0	47.4	52.6	57.0	63.6	69.2	68.5	68.4	65.7	60.2	53.1	48.4		
	2%	DB	50.8	53.6	63.0	71.8	78.4	85.4	87.6	86.8	77.2	67.6	56.4	50.6		
		MCWB	45.8	45.4	49.9	55.1	62.0	67.4	67.9	68.0	63.9	58.9	50.9	46.6		
	5%	DB	47.1	50.3	58.8	67.9	74.4	81.7	83.7	82.6	73.1	63.9	52.9	47.9		
		MCWB	43.2	44.1	48.9	53.9	59.6	65.9	66.8	66.9	62.1	57.1	49.0	44.6		
	10%	DB	44.2	46.6	54.8	63.5	70.4	77.6	79.9	78.5	69.4	60.6	50.2	45.1		
		MCWB	41.2	42.0	46.7	51.8	58.1	64.3	65.4	65.4	60.8	55.6	47.1	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	48.9	54.0	58.6	65.8	71.6	71.6	71.3	67.4	62.1	54.1	49.7		
		MCDB	53.3	55.2	64.4	72.8	79.4	85.8	87.1	85.5	77.7	69.0	59.2	52.5		
	2%	WB	46.3	46.4	51.5	56.3	63.3	68.9	69.3	69.5	65.2	59.9	51.7	47.1		
		MCDB	50.0	52.6	59.8	69.0	75.2	81.8	83.2	82.6	74.4	65.6	55.4	50.1		
	5%	WB	43.5	44.3	49.4	54.4	61.2	67.0	67.7	67.8	63.3	58.0	49.5	44.9		
		MCDB	46.8	49.5	56.9	65.7	71.5	78.7	80.6	79.7	71.3	62.8	52.5	47.6		
	10%	WB	41.2	42.2	47.5	52.5	59.1	65.2	66.3	66.3	61.5	56.1	47.4	42.5		
		MCDB	43.9	46.4	54.1	62.2	68.1	75.7	77.4	76.9	68.1	59.9	49.9	44.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	13.6	17.6	19.8	19.3	19.8	20.0	19.4	17.8	14.7	10.9	9.3		
		MCDBR	14.3	21.3	26.3	29.5	28.0	27.4	27.7	27.5	23.7	20.0	15.7	12.9		
		MCWBR	10.4	13.9	15.1	14.9	13.4	11.6	10.8	11.2	12.1	12.0	10.8	9.5		
	5% WB	MCDBR	13.2	18.8	22.9	26.2	24.6	24.9	24.8	24.2	21.5	18.1	14.6	11.7		
		MCWBR	10.0	12.8	13.6	13.5	12.1	10.9	10.2	10.6	11.3	11.2	10.4	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.322	0.358	0.392	0.405	0.409	0.411	0.404	0.378	0.353	0.320	0.293		
	taud		2.452	2.379	2.332	2.267	2.271	2.300	2.289	2.325	2.377	2.433	2.475	2.457		
	Ebn at Noon		251	266	272	273	273	272	269	266	262	252	239	238		
	Edh at Noon		21	28	34	40	41	40	40	37	32	26	20	18		
(44) All-Sky Solar Radiation	RadAvg		363	628	1012	1372	1585	1817	1754	1532	1147	719	399	298		
	RadStd		39	85	135	196	159	169	161	159	118	78	48	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	N/A
Regional (116 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.61	N/A	N/A	+101	+27

Nomenclature: See separate page